1. Credit Card Processing System

1.1 Introduction

Credit cards are now a staple for everyday purchases, whether online, in-store, or overseas. Every single payment involves a network of different parties—the customer, the merchant, the payment gateway, the acquiring bank, and the issuing bank. Because money is moving in real time, every transaction needs to be checked, cleared, and settled quickly and securely.

A dedicated Credit Card Processing System automates this entire chain of events. By taking manual tasks out of the loop, the system speeds up transactions, reduces human error, catches fraud before it happens, and gives customers, merchants, and banks clear, up-to-second visibility into their finances. It handles everything from issuing new cards and authorizing everyday swipes to settling merchant payouts, sending out monthly bills, and keeping detailed records for business analysis.

1.2 Software Requirements Specification (SRS)

1.2.1 Purpose of this Document

This SRS document outlines the functional and technical requirements for building the Credit Card Processing System. It acts as a shared reference point for software engineers, bank managers, merchants, and end-users, ensuring everyone is aligned on what the system will do, how it will work, and what scope it covers across every phase of development.

1.2.2 Scope of the System

The system manages the full lifecycle of a credit card account. This includes accepting applications, issuing cards, checking and approving purchases in real time, settling accounts, generating statements, collecting payments, and flagging suspicious activity.

- **Estimated Development Cost:** $80,000
- **Target Delivery Timeline:** 8 Months

1.2.3 System Overview

The platform serves as the central hub connecting cardholders, merchants, payment gateways, and banking networks. By automating routine financial workflows, it keeps payment data accurate, secure, and moving without friction. Key modules include:

- **Card & Account Management**
- **Real-time Authorization Engine**
- **Clearing & Settlement Processor**
- **Billing & Statements**
- **Fraud Detection & Risk Alerts**
- **Management Reporting**

1.2.4 General Description

Core Functions

- **Card Lifecycle:** Accept applications, verify applicants, assign credit limits, activate new cards, and block or close accounts when needed.
- **Payment Authorization:** Check card validity, balances, and merchant credentials to approve or decline purchases instantly at POS terminals or online checkouts.
- **Clearing & Settlements:** Process merchant payouts, calculate service fees, and handle customer refunds or disputed charges (chargebacks).
- **Billing & Collections:** Calculate interest and fees, compile monthly statements, and collect payments through various channels.
- **Security & Insights:** Monitor transactions for signs of fraud and generate regular business performance reports for management.

User Objectives

- **Cardholders & Merchants:** Fast, seamless, and dependable payment approvals at checkout.
- **Bank Staff:** Simple tools for updating accounts, reviewing credit limits, and handling customer disputes.
- **Fraud Teams:** Automatic detection tools to spot suspicious activity early and minimize financial losses.
- **Executive Leadership:** Accurate reports on card usage, revenue trends, and account health to guide business decisions.

System Users

- **Cardholders:** Individuals who make purchases, manage card settings, review monthly statements, and pay their bills.
- **Merchants:** Retailers and online stores that accept card payments and receive daily settlements.
- **Bank Staff:** Customer service reps, credit underwriters, fraud analysts, and department managers.
- **System Administrators:** IT staff responsible for maintaining server uptime, managing access rights, and overseeing data security.

Key Benefits

- Faster purchase authorizations and reliable daily settlements.
- Stronger customer trust through proactive security and account visibility.
- Fewer manual input errors and better protection against payment fraud.
- Clear analytics to help bank leadership track performance and profitability.

1.2.5 Functional Requirements

- **Card Management**
    - Process incoming card applications, verify user identity details, and set starting credit limits.
    - Handle card activation, status changes (active, frozen, blocked), card replacements, and account closures.
- **Transaction Processing**

    - Verify card details, available credit, and merchant legitimacy instantly during checkout.
    - Execute real-time approvals, declines, settlements, and partial or full refunds.
    - Keep an audit log of every transaction attempt along with its final status code.
- **Billing & Payments**
    - Calculate monthly purchase totals, accrued interest, standard fees, and taxes automatically.
    - Generate monthly billing statements and accept payments through bank transfers, online portals, or debit channels.
- **Fraud Detection & Prevention**
    - Flag unusual transactions based on purchase size, geographic location, or sudden spikes in frequency.
    - Send immediate SMS/email alerts to cardholders and support instant, temporary card freezes via mobile app or web.
- **Merchant Management**
    - Onboard new merchants, verify their business profiles, and manage their bank payout details.
    - Calculate merchant discount rates (MDR) and deposit net settlement funds into merchant accounts.
- **Reporting & Analytics**
    - Auto-generate daily, weekly, and monthly reports covering transaction volume, gross revenue, overdue accounts, and fraud metrics.

1.2.6 Interface Requirements

- **Software Interfaces**
    - **Card Networks:** Direct API integration with network switches (e.g., Visa, MasterCard) via secure channels.
    - **Core Banking System:** Connects to the bank's central ledger to sync customer accounts, balances, and credit caps.
    - **Messaging Gateways:** SMS and email service APIs for transactional notifications, OTPs, and monthly statements.
    - **Business Intelligence Tools:** Analytics export channels for reporting dashboards.
- **User Interfaces**
    - **Bank Staff Dashboard:** Intuitive interface for managing card accounts, investigating dispute claims, acting on fraud alerts, and pulling reports.
    - **Cardholder Portal & App:** Clean, mobile-friendly interface for checking balances, viewing statements, turning cards on/off, and paying bills.
    - **Merchant Portal:** Simple portal for viewing daily sales summaries, settlement schedules, and chargeback claims.

1.2.7 Performance Requirements

- **Concurrent Capacity:** Process up to 5,000 transactions simultaneously without performance drops.
- **Response Time:** Complete purchase authorization requests within **3 seconds**.
- **System Availability:** Maintain **99.99% overall uptime**.

- **Accuracy:** Keep transaction error rates below **0.05%**.

1.2.8 Design Constraints

- **Hardware Constraints**
  - Dedicated server infrastructure with at least 32 GB RAM, 2 TB SSD storage, and a failover backup server.
  - Hardware Security Modules (HSM) for managing encryption keys and securing PIN/PAN data.
- **Software Constraints**
  - Built using a secure framework (Java or Python).
  - Relational database backend using PostgreSQL, MySQL, or Oracle for ACID-compliant financial logging.
- **Regulatory & Compliance Constraints**
  - Strictly compliant with **PCI DSS** standards for handling credit card data.
  - Fully aligned with **GDPR** (or regional data privacy laws) alongside standard **KYC (Know Your Customer)** and **AML (Anti-Money Laundering)** regulations.

1.2.9 Non-Functional Attributes

- **Security:** Role-based access control (RBAC), end-to-end data encryption, and sensitive card tokenization.
- **Portability:** Web and mobile interfaces compatible across all major browsers and modern operating systems.
- **Reliability:** Automatic server failover and instant transaction rollbacks if a process fails mid-way.
- **Scalability:** Easily handles growth in user volume, higher transaction throughput, and new feature additions over time.
- **Data Integrity:** Strict relational checks to ensure financial records, account balances, and ledgers remain accurate and tamper-proof.

1.2.10 Schedule and Budget

Project Schedule

1. **Requirements Gathering & Analysis:** 1 Month
2. **System Architecture & UI Design:** 2 Months
3. **Core Application Development:** 4 Months
4. **Testing, QA & Final Deployment:** 1 Month

Budget Breakdown

- **Development:** $50,000
- **Testing & Security Audits:** $15,000
- **Deployment & Initial Maintenance:** $15,000
- **Total Estimated Budget: $80,000**
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1.3 Class Diagram:

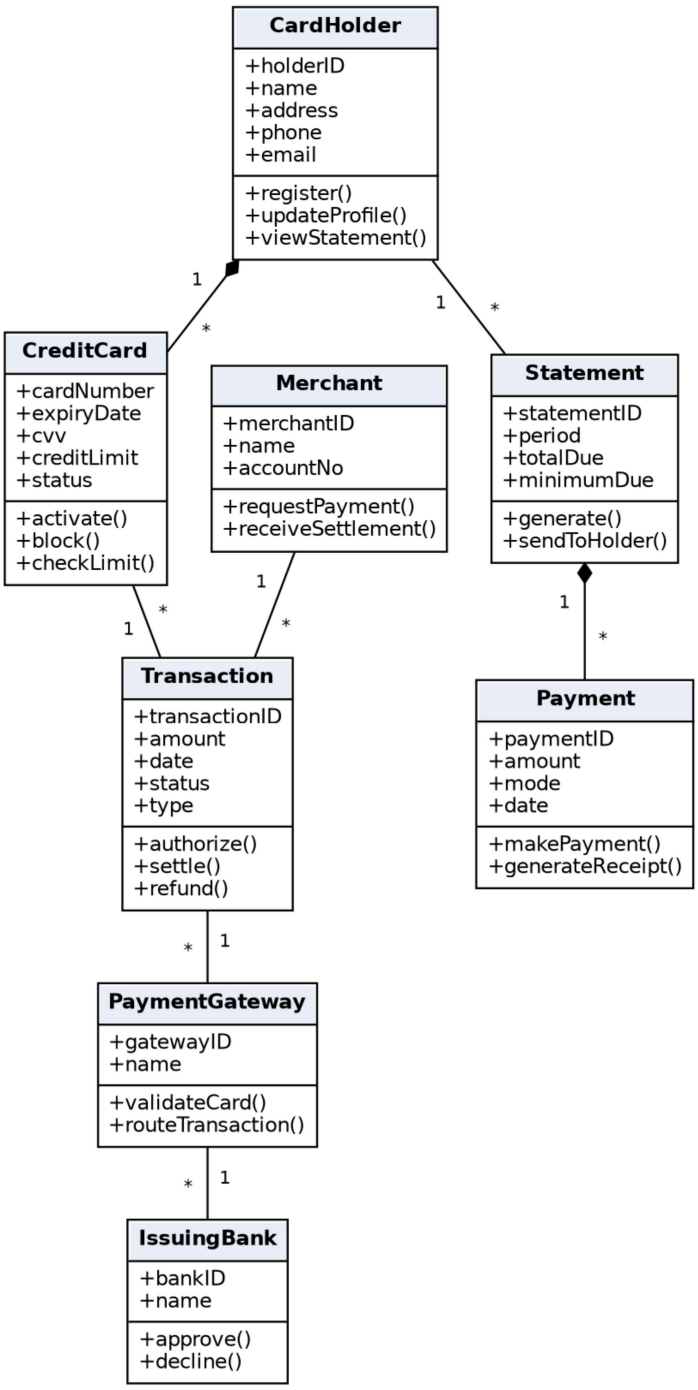


Fig 1.1

Brief Description:

The class diagram for the Credit Card Processing System (CCPS) represents the key entities, their attributes, and relationships in the system. Below are the main components typically included in the class diagram for CCPS:

- **CardHolder:** stores customer details and owns one or more credit cards.

- **CreditCard:** holds card number, expiry date, credit limit and status; every transaction is made against a card.
- **Merchant:** represents a business that requests payments and receives settlements.
- **Transaction:** records each purchase with amount, date, type and status, and supports authorization, settlement and refund.
- **PaymentGateway:** validates card details and routes each transaction to the issuing bank.
- **IssuingBank:** approves or declines transactions based on account status and available credit.
- **Statement:** summarizes card activity for a billing period, including total and minimum amount due.
- **Payment:** records payments made by the cardholder against a statement.